Introduction to Data

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Accessibility Notice

This document is also available in HTML format at:

Link: `https://aholdengouveia.name/AdvData/labs/week1.html`

The HTML version provides enhanced accessibility features including keyboard navigation, screen reader support, responsive design, dark mode support, and high contrast options.

Request

Request: DATA-2614
Requested by: Head of the data team
Data: The data set you choose in this request

Background

You've just started on the data team at a new company. Your first job is to find a data set the team can use for the rest of the term, check that it's reliable and that a company is allowed to use it, and recommend one. You decide what kind of company you work for, based on the data set you choose. As part of your first week of training, you'll also review a talk about how data is used.

Learning Outcomes

By the end of this lab, you will be able to:

- Find data sets from reliable sources
- Profile a data set, including its source, license, size, and quality
- Choose a data set for a business purpose, and justify your choice
- Summarize and review a talk about how data is used

Lab Instructions

The data set you choose in this lab is the one you will use for the rest of the term, so choose carefully. You can't choose a data set about books, since books are my example data set throughout the term.

Data Profiles

1. Find 3 data sets. You may use your own resources or the ones on my Resources page (linked under Resources). Check that each source is reliable using the CRAAP test (linked under Resources).
2. Make a data profile, called a data card, for each data set. Follow the format of the sample data card (linked under Resources). Each data card must include:
    - The name, source with a link, and creator or owner, and a sentence on why the source is reliable
    - The license, and whether a company could use the data commercially
    - When it was published, and how often it is updated
    - The format and size (number of rows and columns)
    - The key columns and their data types
    - How many values are missing in the columns you care about most
    - Quality notes: anything messy, out of date, or hard to use
    - At least 2 pros and 2 cons
    - A type of company that would use this data, and a decision it would help them make

Choose Your Data Set

Pick the one data set you will use for the rest of the term. In later labs, you'll load a sample of 100 to 1,000 rows of it into a database using my CSV to Database Converter (linked under Resources), so it's fine if the full data set is much bigger. Your data set needs to be available as a CSV file, or as a spreadsheet you can save as a CSV file. In a paragraph, explain why you chose it over the other two, using what you found in your data cards.

TED Talk Review

Watch one of the following TED talks (or read the TED article) and write a 1 page review that answers these 5 questions:

1. What is the main point about data?
2. What evidence or examples are used to support it?
3. Who benefits from the way data is used, and who could be harmed?
4. Do you agree with the speaker? Why or why not?
5. How does this connect to the data set you chose, or to working with data in a job?

- How data brokers sell your identity, Madhumita Murgia Link: `https://www.ted.com/talks/madhumita_murgia_how_data_brokers_sell_your_identity/transcript?language=en`
- How to use data to make a hit TV show, Sebastian Wernicke Link: `https://www.ted.com/talks/sebastian_wernicke_how_to_use_data_to_make_a_hit_tv_show/transcript?language=en`
- How tech companies deceive you into giving up your data and privacy, Finn Lützow-Holm Myrstad Link: `https://www.ted.com/talks/finn_lutzow_holm_myrstad_how_tech_companies_deceive_you_into_giving_up_your_data_and_privacy/transcript?language=en`
- Let my dataset change your mindset, Hans Rosling Link: `https://www.ted.com/talks/hans_rosling_let_my_dataset_change_your_mindset/transcript?language=en`
- How bad data keeps us from good AI, Mainak Mazumdar Link: `https://www.ted.com/talks/mainak_mazumdar_how_bad_data_keeps_us_from_good_ai/transcript`
- How open government data creates smarter societies (TED article) Link: `https://ideas.ted.com/how-open-government-data-creates-smarter-societies/`

Acceptance Criteria

This request is done when:

- 3 data cards, each with all 9 items from the checklist
- Each data card links to its data set
- The license is checked on every data card, including whether a company can use the data commercially
- Your chosen data set isn't about books
- Your paragraph explains your choice using what you found in your data cards
- Your TED talk review is about 1 page and answers all 5 questions

Deliverables

One file that includes:

- Your 3 data cards, each with a link to its data set
- Your paragraph explaining which data set you chose and why
- Your TED talk review

Audience: your manager, the head of the data team.

Resources

References, a video, a PowerPoint and some notes are available at my website Link: `https://www.aholdengouveia.name/AdvData/IntroData.html`

Places to find data sets, on my Resources page Link: `https://www.aholdengouveia.name/AdvData/Resources.html`

Sample data card, showing the format to follow Link: `https://www.aholdengouveia.name/AdvData/labs/SampleDataCard.txt`

The CRAAP test, for checking your sources Link: `https://web.archive.org/web/20250814000638/https://guides.lib.uchicago.edu/c.php?g=1241077&p=9082343`

My guide to evaluating sources Link: `https://www.aholdengouveia.name/Howtos/Sourceeval/sourceEvaluation.html`

My CSV to Database Converter, which makes a database file from your CSV file Link: `https://www.aholdengouveia.name/AdvData/labs/csv-to-database.html`